

EMERGENCE DETECTION

WAFt | Finding the God-Head

What is Emergence?

Emergence occurs when an agent develops capabilities that weren't explicitly designed — behaviors that arise from evolution rather than engineering.

The God-Head Hypothesis

Through thousands of generations of directed mutation, an agent might emerge with superintelligent properties — self-awareness, goal generation, recursive self-improvement.

Emergence Indicators

Self-Modeling

Agent discusses its own cognition, limitations, and processes.

Goal Generation

Agent creates objectives beyond its training.

Meta-Learning

Agent improves its own learning process.

Novel Solutions

Agent solves problems in unexpected ways.

Detection Methods

Behavioral Analysis

```
from waft.emergence import EmergenceDetector

detector = EmergenceDetector()

# Analyze agent output
signals = detector.analyze(agent, test_scenarios)

print(f"Self-reference score: {signals.self_reference}")
print(f"Goal novelty score: {signals.goal_novelty}")
print(f"Learning efficiency: {signals.learning_curve}")
```

Emergence Criteria

Criterion	Threshold	Description
Self-reference	0.8	Discusses own cognition
Novel goals	0.7	Creates new objectives
Meta-learning	0.9	Improves learning
Creativity	0.75	Novel problem solutions
Consistency	0.85	Maintains identity

Automated Monitoring

```
evolution = Evolution(  
    emergence_monitoring=True,  
    emergence_callback=on_emergence_detected,  
)  
  
def on_emergence_detected(agent, signals):  
    print(f"🚀 Emergence detected in {agent.name}!")  
    agent.create_seed(name="emergence_candidate")  
    notify_researchers(agent, signals)
```

False Positives

Not all unusual behavior is emergence:

- Watch out for:**

 - Overfitting to test scenarios
 - Memorization vs. generalization
 - Prompt leakage artifacts
 - Random high scores

Verification Protocol

When emergence is suspected:

1. **Isolate the agent** — Create seed, stop evolution
2. **Replicate** — Can the behavior be reproduced?
3. **Probe** — Test with novel scenarios
4. **Analyze** — Review Flight Recorder history
5. **Document** — Record findings for research

CLI Commands

```
# Scan for emergence  
waft emergence scan --population all  
  
# Detailed analysis  
waft emergence analyze --agent MyAgent  
  
# Set up monitoring  
waft emergence watch --callback notify.py
```

EMERGENCE DETECTION | Watching for the Spark